WELCOME TO THE NEW ECONOMY · 2026 EDITION

Web3 Insider: The Financial Infrastructure Handbook

250+ concepts across digital assets, DeFi, tokenisation, compliance, risk and fraud

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Your concise guide to the language of the new digital economy — designed for finance and technology professionals navigating the Web3 landscape.

Anastasiya Kukharuk

CHAPTER 01

Basics

The foundation — concepts without which the rest makes no sense.

— Cryptocurrency

A digital currency secured by cryptography and operating on a decentralised network — without a central bank or issuer. Bitcoin (BTC) is the first and largest; Ether (ETH) is the fuel of the Ethereum network. Not all cryptocurrencies are stablecoins — most are volatile assets.

BASICS

— Fiat Currency

Government-issued currency not backed by a physical commodity — its value derives from government decree and public trust. USD, EUR, GBP are fiat currencies. The term 'fiat' comes from Latin: 'let it be done'. In crypto, fiat = traditional money, as opposed to digital assets.

BASICS

— TradFi (Traditional Finance)

The conventional financial system — banks, exchanges, brokers, asset managers, central banks. Operates on legacy infrastructure with settlement delays, intermediaries and business-hours constraints. Web3 positions itself as an alternative to TradFi rails.

CONCEPT

— CeFi (Centralised Finance)

Crypto services provided by centralised companies — exchanges (Binance, Coinbase), lenders (now defunct Celsius, BlockFi). Offers crypto exposure with a familiar custodial model, but carries counterparty risk. The collapse of FTX (2022) was the defining CeFi failure.

CONCEPT

— DeFi (Decentralised Finance)

An umbrella term for financial protocols and applications built on public blockchains that replicate and extend traditional financial services — lending, trading, derivatives — without centralised intermediaries. Schär (2021, Federal Reserve Bank of St. Louis) defines DeFi as an 'open, permissionless, and highly interoperable protocol stack built on public smart contract platforms'.

BASICS

— Digital Asset

A broad term for any value-bearing asset that exists in digital form on a blockchain or DLT — including cryptocurrencies, stablecoins, security tokens, NFTs and CBDCs. Not all digital assets are cryptocurrencies.

BASICS

— On-Chain / Off-Chain

On-chain = recorded and settled on the blockchain (transparent, immutable). Off-chain = processed outside the blockchain (faster, cheaper, but less transparent). Many real-world applications combine both.

CONCEPT

— Permissioned vs Permissionless Blockchain

Permissionless (public): anyone can join and transact — Bitcoin, Ethereum. Permissioned (private/consortium): access is controlled by a central authority — used by banks and enterprises (e.g. R3 Corda, Hyperledger). Different trade-offs between openness and control.

INFRASTRUCTURE

— Blockchain

A distributed ledger technology (DLT) in which transaction records are grouped into cryptographically linked blocks, forming an append-only chain. Consensus among network participants — rather than a trusted central authority — ensures data integrity. First formalised by Nakamoto (2008) in the Bitcoin white paper.

INFRASTRUCTURE

— Web3

The third era of the internet built on decentralised protocols, where users own their data and assets rather than ceding control to platforms.

CONCEPT

— Decentralisation

A model with no single point of control — decisions are made by a network of nodes or token holders through governance mechanisms.

CONCEPT

— Node

A computer participating in a blockchain network — it verifies transactions, stores a copy of the ledger, and maintains network security.

INFRASTRUCTURE

— Wallet

An application or device that stores cryptographic keys. It does not hold tokens — it holds access to them on the network.

TOOLS

— Private Key / Public Key

A cryptographic key pair: the public key is your address (shareable), the private key is your password — losing it means permanent loss of access to your assets.

SECURITY

— Seed Phrase

12 or 24 words generated when creating a wallet. The master key to all assets — never photograph it digitally or store it in the cloud.

SECURITY

— Consensus Mechanism

The algorithm by which network nodes agree on the state of the ledger. The most important: Proof of Work (PoW) and Proof of Stake (PoS).

INFRASTRUCTURE

— Proof of Work (PoW)

A consensus mechanism where miners compete to solve computational puzzles. Energy-intensive but historically proven — used by Bitcoin.

INFRASTRUCTURE

— Proof of Stake (PoS)

Validators lock (stake) tokens as collateral. More energy-efficient than PoW — used by Ethereum after 'The Merge' (2022).

INFRASTRUCTURE

— Transaction Hash (TxHash)

A unique identifier for every transaction on a network — the equivalent of a transfer reference number, but public and verifiable by anyone on a block explorer.

TOOLS

— Block Explorer

A public search engine for transactions and addresses on a blockchain (e.g. Etherscan for Ethereum). Full transparency.

TOOLS

— Gas Fee

The fee for executing an operation on the Ethereum network — paid to validators in ETH. The more congested the network, the higher the gas.

INFRASTRUCTURE

— Layer 1 (L1)

The base blockchain layer (Bitcoin, Ethereum, Solana). Responsible for final transaction settlement and network security.

INFRASTRUCTURE

— Layer 2 (L2)

A protocol built on top of L1 that processes transactions faster and cheaper, recording results to L1 in batches (e.g. Arbitrum, Optimism, zkSync).

INFRASTRUCTURE

— Interoperability

The ability of different blockchain networks to communicate and exchange assets — a key technical challenge in the Web3 ecosystem.

INFRASTRUCTURE

— Fork

A change to a network's protocol. Soft fork = backwards-compatible update. Hard fork = a network split into two separate chains (e.g. Ethereum / Ethereum Classic).

INFRASTRUCTURE

— Mainnet / Testnet

Mainnet is the production network with real value. Testnet is the development environment with 'fake' tokens of no value.

INFRASTRUCTURE

— Airdrop

A free distribution of tokens to wallets meeting certain criteria — often a reward for early use of a protocol.

TOKENOMICS

— Whitepaper

A technical document describing a project, its mechanisms, tokenomics and vision. The equivalent of a prospectus in the crypto world.

GOVERNANCE

— DAO (Decentralised Autonomous Organisation)

An organisation governed by smart contracts and token holder voting — no board, no CEO, decisions made by the community.

GOVERNANCE

— Tokenomics

The economics of a token: supply, inflation, distribution mechanisms, vesting, burning. Essential analysis before any investment.

CONCEPT

— Hot Wallet / Cold Wallet

Hot wallet: connected to the internet (exchange accounts, mobile/browser wallets) — convenient but exposed to online attacks. Cold wallet: kept offline (hardware wallets, paper wallets) — slower to use but the standard for storing significant holdings. Institutional custody models typically split funds between a small hot wallet for liquidity and the bulk held cold.

SECURITY

— ICO (Initial Coin Offering)

A fundraising method in which a project sells newly created tokens directly to the public, typically in exchange for ETH or BTC, before the product exists. Peaked in 2017–2018; the frequent absence of any underlying asset or working product made ICOs the reference point regulators still cite when explaining why token issuance needed rules (see Howey Test, Security Token).

TYPE

— Validator

A network participant responsible for verifying transactions and proposing new blocks under Proof of Stake, having locked (staked) tokens as a financial guarantee of honest behaviour. Distinct from a Node in that not every node validates; slashing removes a portion of a validator's stake if it acts maliciously or goes offline.

INFRASTRUCTURE

CHAPTER 02

Stablecoins

Stable digital assets — the bridge between traditional finance and Web3.

— Stablecoin

A crypto-asset designed to maintain a stable value relative to a specified asset or basket of assets — typically a fiat currency. The ECB (Bullmann et al., 2019) categorises stablecoins by stabilisation mechanism: fiat-collateralised, crypto-collateralised, and algorithmic. MiCA (2023) formally distinguishes between e-money tokens (EMTs) and asset-referenced tokens (ARTs).

BASICS

— Fiat-backed Stablecoin

Each token is backed by real currency held by a central custodian. Examples: USDC (Circle), USDT (Tether), EURI (Banking Circle).

TYPE

— USDT (Tether)

The largest stablecoin by market cap — issued by Tether Ltd, backed 1:1 by USD reserves. Available on 15+ blockchains. Dominant in emerging markets and crypto trading. Controversial due to historical opacity around reserves; now publishes quarterly attestations. Not MiCA-compliant — Tether has not obtained an EU EMI licence, making USDT non-compliant for EU CASPs from 2025.

TYPE

— USDC (USD Coin)

A USD-backed stablecoin issued by Circle — fully regulated, monthly reserve attestations by Grant Thornton. MiCA-compliant (Circle holds an EU EMI licence). The preferred stablecoin for institutional DeFi and on-chain settlements. Available on Ethereum, Solana, Base and other major chains.

TYPE

— USDG (Global Dollar)

A USD-backed stablecoin issued by Paxos under the Global Dollar Network — a consortium model where participating institutions (Robinhood, Kraken, Anchorage) share yield from reserves. Regulated by MAS (Singapore). Designed for institutional payments and cross-border settlement.

TYPE

— EURI (Eurite — Banking Circle)

Banking Circle's EUR-backed stablecoin — the first MiCA-regulated stablecoin issued and backed by a bank in the EU (launched 26 August 2024). 1:1 EUR-pegged e-money token, fully backed, redeemable at par. Available on Ethereum and BNB Smart Chain. Reserve attestations by EY; smart contract code audited by PeckShield.

TYPE

— Crypto-backed Stablecoin

Secured by other crypto assets, usually with overcollateralisation. Example: DAI from MakerDAO — backed by ETH.

TYPE

— Algorithmic Stablecoin

Maintains its peg through algorithms controlling supply, without direct collateral. High risk — spectacular failures (Terra/LUNA 2022).

TYPE

— CBDC (Central Bank Digital Currency)

A digital currency issued by a central bank. The equivalent of cash in digital form, but centralised. Many countries are in pilot phase.

POLICY

— Peg

The mechanism maintaining a token's rate at the target level. 'Depeg' = loss of stability — one of the biggest risks in DeFi.

MECHANISM

— Overcollateralisation

Securing a loan with assets of higher value than the borrowed amount (e.g. 150%). Protects against fluctuations in the value of underlying assets.

MECHANISM

— Mint / Burn

Mint = creating new tokens (e.g. you deposit $1, you receive 1 USDC). Burn = destroying tokens on redemption (you buy back USDC for $1, the token disappears).

MECHANISM

— Reserve Attestation

A public report confirming that a stablecoin issuer actually holds the declared reserves. A key element of transparency.

COMPLIANCE

— Yield-bearing Stablecoin

A stablecoin generating automatic returns for the holder (e.g. sDAI, USDY). A revolution in corporate liquidity management.

INNOVATION

— Yield-Bearing Token

A token that automatically accrues returns to its holder — either through rebasing (balance increases) or appreciation (value increases). Examples: stETH (liquid staking), sDAI (savings DAI), tokenised money market funds. Used by treasuries for on-chain yield without manual reinvestment.

INNOVATION

— Payment Stablecoin

A stablecoin designed strictly for payments — fast, cheap, without native yield. A model supported by EU regulators (MiCA) and, since 2025, by the US GENIUS Act (see Chapter 07).

TYPE

— E-Money Token (EMT)

MiCA's regulatory category for stablecoins backed by a single fiat currency. Issuers must hold an e-money institution licence.

COMPLIANCE

— Asset-referenced Token (ART)

MiCA's category for tokens stabilised by a basket of assets (e.g. currencies, commodities). Stricter capital requirements than EMTs.

COMPLIANCE

— Liquidity Pool

A pool of stablecoins (or other assets) locked in a smart contract, enabling trading without a traditional order book.

DEFI

— Curve Finance / 3Pool

The leading protocol for swapping stablecoins (DAI/USDC/USDT) with minimal slippage. Key DeFi infrastructure.

DEFI

— Real-Time Gross Settlement on-chain

The concept of real-time gross settlement executed on a blockchain — stablecoins as a settlement instrument between banks.

INNOVATION

— Cross-border Stablecoin Payments

International transfers via stablecoins: settlement in seconds, cost a fraction of a cent vs. traditional SWIFT (1–5 days, high fees).

USE-CASE

— Depegging Risk

The risk of a stablecoin losing its target peg. Can be triggered by a loss of confidence, a speculative attack, or problems with reserves.

RISK

— Programmable Money

Money with built-in business logic — e.g. automatic salary payment at midnight, or a payment released only upon fulfilment of conditions.

INNOVATION

— Correspondent Banking vs Stablecoins

Traditional transfers use a network of correspondent banks (slow, expensive). Stablecoins replace this infrastructure with a single blockchain network.

CONCEPT

CHAPTER 03

Tokenization

Moving real-world assets onto blockchain — the biggest institutional trend.

— Tokenized Financial Instruments

Traditional financial instruments — bonds, equities, funds, derivatives — represented as tokens on a blockchain. Enables fractional ownership, T+0 settlement, 24/7 trading and programmable compliance. Regulated as securities in most jurisdictions. Examples: BlackRock BUIDL (tokenised money market fund), UBS tokenised bonds.

TYPE

— Tokenization

The process of representing rights to an asset (real estate, shares, bonds, artwork) as a token on a blockchain. Each token = a fraction of ownership.

BASICS

— RWA (Real-World Assets)

Real-world assets moved onto blockchain: government bonds, real estate, private equity, commodities. A market estimated at trillions of dollars.

ASSET-CLASS

— Security Token

A token representing a financial instrument (share, bond). Subject to regulation like traditional securities — requires licencing and KYC.

TYPE

— Howey Test

The US Supreme Court test (SEC v. W.J. Howey Co., 1946) used to determine whether an asset is an 'investment contract' — and therefore a security under US law: an investment of money, in a common enterprise, with an expectation of profit derived from the efforts of others. The default reference point used by the SEC and courts to decide whether a given token is a security token.

REGULATION

— Utility Token

A token giving access to a platform's services or product — does not represent ownership. Usually not a security (depends on the regulator).

TYPE

— Fractionalization

Splitting a high-value asset into small tokens — e.g. a $1M property divided into 1,000,000 tokens at $1 each. Democratisation of investment.

MECHANISM

— Token Standard

A technical specification defining a token's behaviour on the network. Most important: ERC-20 (fungible), ERC-721 (NFT), ERC-1400 (security tokens).

TECHNICAL

— ERC-20

The standard for fungible tokens on Ethereum. Each token is identical to another of the same type — currency, governance token, stablecoin.

TECHNICAL

— ERC-721

The standard for non-fungible tokens (NFTs) on Ethereum. Each token is unique — digital ownership, certificates, identity documents.

TECHNICAL

— ERC-1400 / ERC-3643

Standards for security tokens with built-in compliance mechanisms (whitelists, transfers requiring authorisation, KYC on-chain).

TECHNICAL

— Cap Table on Blockchain

A shareholder register managed by a smart contract — automatic updates on transfers, instant transparency for the company.

USE-CASE

— Tokenised Treasury Bonds

Government bonds (e.g. US T-Bills) represented as tokens — high liquidity, instant settlement, 24/7 availability. BlackRock BUIDL as an example.

USE-CASE

— Tokenised Real Estate

Tokenised real estate enables fractional ownership and global access to property markets without intermediaries and barriers to entry.

USE-CASE

— Digital Twin

A digital representation of a physical asset on the blockchain — synchronised with its real-world state through off-chain data.

CONCEPT

— Oracles

Services providing real-world data (prices, events, market data) to smart contracts. Critical for RWA tokenisation. Chainlink is the market leader.

INFRASTRUCTURE

— Digital Asset Custody

Secure storage of private keys to tokens — by a licensed custodian (e.g. Fireblocks, Anchorage). A regulatory requirement for institutions.

COMPLIANCE

— Transfer Restrictions

Transfer restrictions built into a token — e.g. only to verified investors, only in certain jurisdictions. Required for security tokens.

COMPLIANCE

— Primary Issuance

The first issue of tokens by the issuer — the equivalent of an IPO or bond issue, but on the blockchain. A cheaper and faster process.

PROCESS

— Secondary Market for Tokens

The secondary market for security tokens — alternative trading platforms (ATS, MTF) enabling trading after issuance.

MARKET

— T+0 Settlement

Settlement of a transaction on the same day (or a few seconds after). Tokenisation enables T+0 vs traditional T+2 in capital markets.

INNOVATION

— Tokenization Platform

Infrastructure for issuing and managing tokens: Securitize, DigiShares, SDX (SIX Digital Exchange), DTCC Digital Assets.

TOOLS

— Lock-up / Vesting

A period during which tokens cannot be sold — protects against sudden sell-offs. Typical for employee and early-stage investor issuances.

MECHANISM

— Spot Bitcoin ETF

An exchange-traded fund holding actual Bitcoin, giving investors price exposure through a regular brokerage account without self-custody. Approved by the US SEC in January 2024 (effective 11 January 2024) after more than a decade of rejected applications, in filings from issuers including BlackRock and Fidelity — the first large-scale TradFi wrapper for direct spot crypto exposure.

USE-CASE

— Spot Ethereum ETF

The equivalent TradFi wrapper for direct spot Ether exposure. The SEC approved the underlying exchange rule changes in May 2024, with the funds themselves beginning trading in July 2024; most approved products do not pass through staking rewards to holders.

USE-CASE

CHAPTER 04

DeFi & Infrastructure

Decentralised finance — protocols replacing traditional financial institutions.

— DeFi (Decentralised Finance)

An ecosystem of financial protocols operating on smart contracts without intermediaries. Lending, trading, derivatives — all peer-to-peer.

BASICS

— Smart Contract

Self-executing code deployed on a blockchain that automatically enforces predefined contractual terms when specified conditions are met — without reliance on a trusted intermediary. Introduced by Nick Szabo (1994) and operationalised by the Ethereum protocol (Wood, 2014).

INFRASTRUCTURE

— CEX (Centralised Exchange)

A crypto exchange with a known operator — company, licence, order book, custody of user funds. Examples: Coinbase, Binance, Kraken. Subject to KYC/AML requirements as a CASP under MiCA.

INFRASTRUCTURE

— DEX (Decentralised Exchange)

An exchange without a central operator — trading directly between wallets via smart contracts. Uniswap, Curve, dYdX are the largest examples.

PROTOCOL

— DEX with Known Operator

A DEX where the underlying smart contracts are deployed and maintained by an identifiable legal entity (e.g. Uniswap Labs). The front-end interface may be regulated even if the protocol itself is permissionless — a growing area of MiCA and FATF scrutiny.

COMPLIANCE

— Permissionless Protocol

A DEX or DeFi protocol with no identifiable operator — fully autonomous smart contracts, no company, no off switch. Regulators face a fundamental challenge: there is no entity to licence, sanction or serve a Travel Rule obligation upon.

COMPLIANCE

— AMM (Automated Market Maker)

A class of decentralised exchange protocol that uses a mathematical pricing function — rather than a traditional order book — to determine asset prices based on pool composition. The constant product formula ($x \times y = k$), introduced by Uniswap (Adams et al., 2021), is the dominant AMM model. Price impact (slippage) increases non-linearly with trade size relative to pool depth.

MECHANISM

— Liquidity Pool

A pool of two or more tokens locked in a smart contract, enabling anyone to trade against it without a counterparty. LPs deposit equal value of both tokens and earn a share of trading fees. The backbone of DEX trading.

INFRASTRUCTURE

— Liquidity Provider (LP)

A user supplying tokens to a DEX liquidity pool in exchange for a share of transaction fees. Carries impermanent loss risk — the value of the deposited assets can diverge from simply holding them.

ROLES

— Yield Farming

A strategy to maximise returns by actively moving assets between DeFi protocols offering the highest yields — combining LP fees, governance token rewards and compounding. High returns typically signal high risk.

STRATEGY

— APY / APR

APR = annual percentage rate (without compounding). APY = including compounding. In DeFi, APY can be astronomical — and usually signals astronomical risk.

METRICS

— TVL (Total Value Locked)

The total value of assets locked in a DeFi protocol. A key metric for assessing the scale and popularity of a project.

METRICS

— Lending Protocol

A protocol enabling the borrowing and depositing of crypto assets — Aave, Compound. Interest rates set algorithmically by supply and demand.

PROTOCOL

— Collateral

Security provided when borrowing in DeFi. Usually crypto — if the value drops, the protocol automatically liquidates it.

MECHANISM

— Liquidation

Automatic sale of collateral when its value falls below a threshold. Protection for the protocol — painful for the user.

MECHANISM

— Flash Loan

An unsecured loan that must be repaid within the same transaction. Legitimate uses (arbitrage) and DeFi attack vectors.

INNOVATION

— Bridge

A protocol enabling the transfer of assets between different blockchains. One of the most frequently attacked elements of Web3 infrastructure.

INFRASTRUCTURE

— Wrapped Token

A token representing an asset from another network — e.g. WBTC is Bitcoin on Ethereum. Enables the use of BTC in DeFi protocols.

TECHNICAL

— MEV (Maximal Extractable Value)

The profit a validator can extract by manipulating the order of transactions in a block. An ethical and technical challenge in the ecosystem.

ADVANCED

— Gas Optimisation

Techniques to minimise transaction costs through more efficient smart contract code or by choosing the right moment for a transaction.

TECHNICAL

— Staking

Locking tokens in a network or protocol to validate transactions or receive rewards. Passive income from crypto assets.

MECHANISM

— Liquid Staking

Staking while retaining liquidity — instead of frozen tokens you receive a derivative token (e.g. stETH) that you can trade or use in DeFi.

INNOVATION

— Protocol Revenue

Revenue generated by a DeFi protocol from transaction fees — distributed to governance token holders or to the treasury.

ECONOMICS

— Composability (Money Lego)

The ability to combine DeFi protocols like building blocks — the output of one is the input of another. Drives innovation but multiplies risk.

CONCEPT

— Slippage

The difference between the expected and actual price of a transaction on a DEX, resulting from price movement during execution. Critical for large trades.

RISK

— Impermanent Loss

The temporary loss in value experienced by a Liquidity Provider when the price ratio of the two pooled tokens diverges from the ratio at the time of deposit, compared to simply holding the tokens outright. Becomes permanent only once the LP withdraws; can be offset, but not always fully covered, by trading fees earned.

RISK

— Perpetual Futures (Perps)

A derivatives contract that tracks an underlying asset's price with no expiry date, kept aligned to spot price through periodic funding-rate payments between long and short holders. The dominant instrument by trading volume on both centralised and decentralised crypto derivatives venues, and a major source of cascading liquidations in stress events.

PROTOCOL

CHAPTER 05

Identity & Ownership

Digital identity, asset ownership and the future of personal data in Web3.

— NFT (Non-Fungible Token)

A unique cryptographic token — each is distinguishable from others. Can represent: artwork, documents, certificates, tickets, property rights.

BASICS

— Digital Ownership

The concept of true digital ownership — a token in a wallet as proof of ownership, with no possibility of removal by a platform.

CONCEPT

— Self-Custody

Independently storing private keys — without intermediaries. Absolute control and absolute responsibility: 'Not your keys, not your coins'.

SECURITY

— Custodial vs Non-Custodial Wallet

Custodial = the platform holds your keys (like a bank). Non-custodial = you hold the keys. Different models of security and responsibility.

SECURITY

— Hardware Wallet

A physical device (Ledger, Trezor) storing keys offline — the highest level of security for significant assets.

SECURITY

— DID (Decentralised Identifier)

A W3C standard for decentralised digital identity — a unique identifier controlled by its owner, without a central database.

IDENTITY

— Self-Sovereign Identity (SSI)

A model in which an individual has full control over their digital identity and decides to whom and what data they disclose.

IDENTITY

— Verifiable Credential (VC)

A digital document (diploma, KYC certificate, licence) issued by a trusted institution, cryptographically signed and verifiable by anyone.

IDENTITY

— KYC On-Chain

Identity verification encoded as token attributes or a credential — a one-time verification, reusable across different protocols.

COMPLIANCE

— Soulbound Token (SBT)

An NFT that cannot be transferred — permanently bound to a wallet. Possible uses: diplomas, reputation, credit history.

INNOVATION

— ENS (Ethereum Name Service)

A domain name system for Ethereum addresses — vitalik.eth instead of 0x...1a2b. Simplifies identification on the network.

IDENTITY

— Account Abstraction (EIP-4337)

An Ethereum standard enabling 'smart wallets' — social recovery, paying gas in other tokens, transaction limits. The future of Web3 UX.

INNOVATION

— Multi-Sig Wallet

A wallet requiring signatures from multiple private keys — e.g. 2 of 3. Used by organisations for the security of treasury funds.

SECURITY

— On-chain Reputation

A history of transactions and protocol interactions as a form of reputation — an alternative to a credit score in a world without central identity.

CONCEPT

— Digital Asset Inheritance

Planning the transfer of crypto assets after an owner's death — the absence of a cryptographic will = permanent loss of access.

USE-CASE

— Proof of Humanity

A protocol enabling proof that a blockchain address belongs to a real human — a basic component of UBI and governance.

IDENTITY

— Token-Gating

Access to content, communities or events conditional on holding a specific token or NFT. A new model for monetisation and community building.

USE-CASE

— On-chain Certificate

A certificate (course, participation, verification) issued as an NFT or SBT — permanent, verifiable, impossible to forge.

USE-CASE

— Portable Identity

A digital identity not tied to a single platform — you carry your profile, reputation and data between applications. The antithesis of the Facebook model.

CONCEPT

— zkProof Identity

A zero-knowledge proof allowing you to prove a fact (e.g. 'I am over 18') without revealing personal data. Privacy by design.

PRIVACY

CHAPTER 06

Threat & Intelligence

Cybersecurity, attack vectors and analytical tools in the Web3 ecosystem.

— Rug Pull

A scam in which project founders suddenly withdraw all funds from a liquidity pool and disappear. The most common form of fraud in DeFi and NFTs.

SCAM

— Phishing (Crypto)

Attacks impersonating known platforms to steal private keys or seed phrases. Often via fake websites and Discord servers.

THREAT

— Smart Contract Exploit

Exploiting a bug in smart contract code to steal funds. Irreversible — code doesn't lie, but it can have vulnerabilities.

THREAT

— Re-entrancy Attack

A class of smart contract attacks where a malicious contract recursively calls a withdrawal function before the balance is updated. The DAO hack (2016) — a classic example.

THREAT

— Flash Loan Attack

Using flash loans to manipulate prices in DeFi protocols — borrow an enormous sum, manipulate the market, repay within the same transaction.

THREAT

— Bridge Exploit

Attacks on bridge protocols — among the most expensive hacks in Web3 history (Ronin Bridge, March 2022: approx. $600–625M; Wormhole, February 2022: approx. $320M). Bridges are infrastructure honeypots.

THREAT

— Wallet Drainer

A malicious smart contract that, after a victim signs a transaction, empties their wallet. Often hidden in NFT minting links.

THREAT

— Address Poisoning

An attack involving sending a small transaction from an address similar to one of your contacts — counting on you mistakenly copying the wrong address.

THREAT

— On-chain Analytics

Analysis of transaction patterns on a blockchain to identify entities, fund flows and suspicious activity. Blockchain Forensics.

INTELLIGENCE

— Chainalysis / Elliptic / TRM Labs

Leading blockchain intelligence firms providing tools for tracking transactions, identifying addresses and compliance for institutions and regulators.

TOOLS

— Address Clustering

A technique linking multiple blockchain addresses to a single entity based on transaction patterns — key in investigations.

INTELLIGENCE

— Mixer / Tumbler

Services obfuscating the flow of funds by 'mixing' transactions from multiple users. Tornado Cash — an example of OFAC sanctions being applied.

THREAT

— Privacy Coin

A cryptocurrency engineered to obscure sender, receiver and/or amount by default — Monero (ring signatures, stealth addresses) and Zcash (zk-SNARKs, optional shielded transactions) are the leading examples. Flagged as elevated risk by FATF guidance; several EU CASPs and exchanges have delisted privacy coins ahead of MiCA/AMLR implementation because on-chain monitoring tools cannot reliably trace them.

THREAT

— Dusting Attack

Sending micro-transactions (dust) to wallets in order to de-anonymise the owner by analysing subsequent movements of those tokens.

THREAT

— Sandwich Attack

Front-running: a bot detects a pending transaction, inserts its own before and after it, profiting from the price movement caused by the victim.

THREAT

— Exploit Post-Mortem

A public analysis conducted by a project after an attack — describing the attack vector, losses and remedial actions. A transparency standard in DeFi.

PROCESS

— Bug Bounty

A reward programme for reporting security vulnerabilities by external researchers. Ethical hacking as a defensive layer for a protocol.

SECURITY

— Smart Contract Audit

A professional review of smart contract code by security firms (Trail of Bits, OpenZeppelin, Certik). Necessary, but does not guarantee security.

SECURITY

— Social Engineering

Manipulation of project staff or owners to gain access — often more effective than a technical attack.

THREAT

— Fake Token

A token creating the illusion of value through a name identical to a real project. Verifying the contract address is critical before any purchase.

SCAM

— OSINT On-chain

Open Source Intelligence based on blockchain data — identifying entities through analysis of the public transaction ledger.

INTELLIGENCE

— Honeypot Contract

A token contract engineered so that buying works normally but selling is blocked or heavily taxed for anyone except the deployer, trapping buyers' funds. Detectable in advance by simulating a sell transaction before purchase — a standard due-diligence step that most retail buyers skip.

SCAM

— Approval Phishing / Token Approval Exploit

An attack in which a victim is tricked into signing a token 'approval' transaction that grants a malicious contract or address unlimited spending rights over a specific token in their wallet, rather than transferring funds directly. The theft can then occur at any later time without a further signature — distinguishing it from a one-off Wallet Drainer transaction.

THREAT

CHAPTER 07

Compliance

Regulations, standards and legal requirements shaping institutional Web3.

— MiCA (Markets in Crypto-Assets)

An EU regulation (2023/1114), in force in stages from 2024/2025 — the first comprehensive regulation of crypto-assets in Europe. Covers issuers and service providers.

REGULATION

— CASP (Crypto-Asset Service Provider)

MiCA's regulatory category — exchanges, custodians, crypto advisers requiring an EU licence. The equivalent of FATF's VASP.

REGULATION

— VASP (Virtual Asset Service Provider)

The FATF term for entities providing virtual asset services. Subject to AML/CFT requirements.

REGULATION

— AML (Anti-Money Laundering)

Combating money laundering — a set of procedures for identifying customers, monitoring transactions and reporting suspicious activity.

COMPLIANCE

— CFT (Countering Financing of Terrorism)

Countering the financing of terrorism — complementary to AML. Together they form AML/CFT as the compliance standard.

COMPLIANCE

— KYC (Know Your Customer)

Verification of a customer's identity before establishing a business relationship — a regulatory requirement for all VASPs/CASPs.

PROCESS

— KYB (Know Your Business)

Verification of corporate entities — UBO, ownership structure, country of registration. More complex than individual KYC.

PROCESS

— CDD / EDD (Customer Due Diligence)

CDD = standard customer verification. EDD (Enhanced) = in-depth for high risk — PEPs, high volumes, elevated-risk jurisdictions.

PROCESS

— Travel Rule

A FATF (and EU Transfer of Funds Regulation) requirement to pass sender and recipient data for crypto transfers. Under the EU TFR the threshold was removed entirely; FATF Recommendation 16 originally referenced a USD/EUR 1,000 de minimis, which many jurisdictions outside the EU still apply.

REGULATION

— Sunrise Problem

The practical Travel Rule compliance gap that arises because different jurisdictions implement FATF Recommendation 16 on different timelines: a compliant VASP may be legally required to collect and transmit originator/beneficiary data to a counterparty VASP that has not yet implemented equivalent obligations, or that uses an incompatible messaging protocol. A persistent operational headache for cross-border crypto transfers.

CHALLENGE

— PEP (Politically Exposed Person)

Persons holding or having held prominent public positions — require EDD and enhanced monitoring.

RISK

— Sanctions Screening

Verification of customers and transactions against sanctions lists (OFAC, EU, UN). A real-time obligation for VASPs/CASPs.

COMPLIANCE

— OFAC (Office of Foreign Assets Control)

The US agency managing financial sanctions — the SDN list. Sanctions have extraterritorial effect, affecting anyone using USD.

REGULATION

— SAR (Suspicious Activity Report)

A suspicious activity report filed with the financial intelligence unit (FIU). A legal obligation for financial institutions.

COMPLIANCE

— STR (Suspicious Transaction Report)

The European equivalent of a SAR — a suspicious transaction report. An AML preventive-framework requirement for obliged entities.

COMPLIANCE

— AMLD (Anti-Money Laundering Directives)

The EU's preventive AML framework for obliged entities has developed through successive directives (4AMLD, 5AMLD). A separate directive, 6AMLD (2018/1673), deals with the criminal-law definition and sanctioning of money laundering offences rather than firms' preventive obligations. From 2027 the directly applicable AMLR (see below) replaces the preventive directives with a single EU rulebook.

REGULATION

— FATF (Financial Action Task Force)

An intergovernmental organisation setting global AML/CFT standards. 40 Recommendations + Recommendation 15 on virtual assets and Recommendation 16 on the Travel Rule.

REGULATION

— Blockchain Analytics in Compliance

Tools (Chainalysis KYT, Elliptic Navigator) for monitoring blockchain transactions for AML risk — risk scoring per address and transaction.

TOOLS

— KYT (Know Your Transaction)

Transaction-level monitoring using blockchain analytics — analysing the origin, destination and risk profile of each on-chain transfer. Complements KYC as a continuous AML layer.

PROCESS

— KYW (Know Your Wallet)

Due diligence applied to a specific wallet address — assessing its risk exposure based on transaction history, links to sanctioned entities, mixers or darknet markets. Used by CASPs to screen counterparties.

PROCESS

— Risk-Based Approach (RBA)

The principle of proportionality in compliance — the intensity of control measures is adapted to the level of risk posed by the customer and transaction.

PRINCIPLE

— DeFi Compliance Challenge

DeFi without a central operator makes applying KYC/AML requirements difficult. Regulators (MiCA, FATF) are actively seeking solutions for decentralised protocols.

CHALLENGE

— DEX with Identified Operator

A decentralised exchange where a legal entity develops, deploys or controls the protocol — MiCA classifies this as a CASP requiring authorisation. The operator's identifiability triggers full AML/KYC obligations, Travel Rule compliance and regulatory accountability.

REGULATION

— DEX with No Identifiable Operator

A fully autonomous DEX governed only by smart contracts and token holder votes, with no controlling entity. MiCA currently does not regulate these — creating a compliance gap. FATF and EU regulators are actively debating how to apply AML obligations to 'pure' DeFi protocols.

REGULATION

— Fully Decentralised Protocol (Regulatory Grey Zone)

A protocol where governance is distributed among thousands of token holders with no single controller. Regulators globally struggle to assign compliance obligations — if no entity 'provides' the service, traditional VASP/CASP frameworks do not apply. A major open question for Web3 regulation.

CHALLENGE

— DLT Pilot Regime (EU Regulation 2022/858)

An EU sandbox regulation allowing trading and settlement of tokenised financial instruments on DLT infrastructure — without full MiFID II compliance. Active from March 2023, runs until 2026 with possible extension. Permits DLT MTFs, DLT SFs (settlement facilities) and DLT TSS (combined trading + settlement) to operate.

REGULATION

— DLT MTF (Multilateral Trading Facility)

A regulated trading venue for tokenised securities under the DLT Pilot Regime — the blockchain equivalent of a traditional MTF. Must be operated by a licensed investment firm or market operator.

REGULATION

— TFR (Transfer of Funds Regulation)

EU Regulation 2023/1113 — extends the Travel Rule to all crypto-asset transfers with no minimum threshold. In force from December 2024. Unhosted wallets require additional originator/beneficiary information.

REGULATION

— AMLR (Anti-Money Laundering Regulation)

A directly applicable EU regulation replacing the national preventive AML directives — creating a single AML rulebook across all member states from 2027. Explicitly covers CASPs. Introduces a EUR 10,000 cash transaction limit and stricter UBO requirements.

REGULATION

— AMLA (Anti-Money Laundering Authority)

A new EU supervisory authority based in Frankfurt — will directly supervise the highest-risk obliged entities including CASPs with cross-border operations. Operational from 2025/2026.

REGULATION

— GENIUS Act (US)

The Guiding and Establishing National Innovation for US Stablecoins Act — signed into law on 18 July 2025, the first comprehensive federal framework for US payment stablecoins. Requires 100% reserve backing in cash/short-dated Treasuries, subjects issuers to the Bank Secrecy Act (full AML/sanctions programmes), bars non-financial public companies from issuing without approval, and gives stablecoin holders priority in an issuer's insolvency. The US counterpart to MiCA's EMT regime, though narrower in scope (payment stablecoins only).

REGULATION

— eIDAS 2.0 (EU Digital Identity Framework)

An updated EU regulation establishing a European Digital Identity Wallet — enabling citizens to share verified credentials (KYC data, qualifications) across borders. Intersects with Web3 SSI and on-chain KYC models.

REGULATION

— MiFID II and Crypto-Assets

The Markets in Financial Instruments Directive applies to crypto-assets qualifying as financial instruments (security tokens, derivatives). MiCA explicitly excludes assets already covered by MiFID II — the boundary between the two frameworks is a key compliance question.

REGULATION

— ESMA Crypto Guidelines

Technical standards and guidance issued by the European Securities and Markets Authority on MiCA implementation — covering classification of crypto-assets, reverse solicitation, disclosure requirements and market abuse under MiCA.

REGULATION

— Proof of Reserves (PoR)

Cryptographic proof that an exchange or issuer actually holds customer assets — a lesson from the collapse of FTX. A transparency standard.

COMPLIANCE

— DORA (Digital Operational Resilience Act)

An EU regulation in force from 2025 — cyber-resilience of financial institutions. Covers ICT providers, including Web3 firms serving institutions.

REGULATION

— UBO (Ultimate Beneficial Owner)

The natural person(s) who ultimately own or control a legal entity, directly or through a chain of ownership, typically identified via a shareholding or voting-rights threshold (commonly 25%). Identifying the UBO is the core objective of KYB; AMLR tightens EU-wide UBO disclosure and verification requirements.

PROCESS

— Market Abuse Regulation under MiCA (Title VI)

MiCA Articles 86–92, in force since 30 December 2024, prohibiting insider dealing, unlawful disclosure of inside information and market manipulation in crypto-assets admitted to trading. Modelled on the traditional Market Abuse Regulation (596/2014) but adapted for 24/7, pseudonymous, cross-venue crypto markets; applies to any person professionally arranging or executing crypto-asset transactions, not only CASPs.

REGULATION

— Wash Trading

Simultaneously or near-simultaneously buying and selling the same asset to create a misleading appearance of trading volume or price activity, without any genuine change in beneficial ownership. A chronic problem on unregulated crypto exchanges and NFT marketplaces used to inflate rankings or manipulate token metrics; explicitly captured under MiCA's market manipulation prohibition.

THREAT

— Spoofing / Layering

Placing, then cancelling before execution, large orders on one side of an order book to create a false impression of supply or demand, luring other participants into trading at a manipulated price. A form of market manipulation under MiCA Title VI, mirroring the equivalent prohibition in traditional securities markets.

THREAT

— Insider Dealing (Crypto-Assets)

Trading a crypto-asset while in possession of inside information not yet made public — e.g. an unannounced exchange listing or a project's undisclosed security vulnerability — or unlawfully disclosing that information to another person. Prohibited under MiCA Article 89, extending an established TradFi concept to token issuers, CASPs and their staff.

REGULATION

— Qualified Custodian

A US regulatory status (under the Investment Advisers Act and related SEC guidance) required for entities holding client funds or securities on behalf of registered investment advisers. A recurring point of friction in US crypto custody, since it is not always clear which crypto custody arrangements meet the standard — distinct from Digital Asset Custody, which describes the technical/operational practice rather than the legal status.

REGULATION

CHAPTER 08

FX & Liquidity in Web3

Foreign exchange settlement on-chain, liquidity fragmentation and the future of cross-currency finance.

— FX Settlement On-Chain

The execution and settlement of foreign exchange transactions directly on a blockchain — eliminating correspondent banks, CLS delays and T+2 cycles. Atomic settlement in seconds.

INNOVATION

— Atomic Swap

A mechanism enabling the simultaneous exchange of two different assets across blockchains without a trusted intermediary. If one leg fails, the entire transaction reverts.

MECHANISM

— CLS (Continuous Linked Settlement)

The traditional FX settlement system used by major banks — processes trillions daily but operates on T+1/T+2. On-chain alternatives aim to replace it with real-time atomic settlement.

INFRASTRUCTURE

— FX Stablecoin

A stablecoin pegged to a non-USD currency — enabling on-chain multi-currency payments and FX conversion. EURI (Banking Circle, EUR-pegged) is the first bank-backed MiCA-compliant euro stablecoin in the EU.

TYPE

— Cross-Currency Pool

A liquidity pool containing two or more different-currency stablecoins (e.g. USDC/EURI). Enables FX conversion without a broker — price set by AMM algorithm.

DEFI

— FX Slippage On-Chain

Price impact when executing a large FX trade in a cross-currency pool — the larger the order relative to pool depth, the worse the execution rate. Critical for institutional volumes.

RISK

— Liquidity Fragmentation

The dispersion of liquidity across multiple blockchains, Layer 2s and DEXs — making it difficult to achieve tight spreads and efficient execution for large FX orders.

CHALLENGE

— Thin Market Risk

The risk of executing at poor rates due to insufficient liquidity depth in cross-currency pools — particularly acute for non-USD pairs in early-stage Web3 FX markets.

RISK

— PvP (Payment versus Payment)

The simultaneous exchange of two payment legs — the on-chain equivalent of CLS. Eliminates the FX settlement risk (Herstatt risk) inherent in sequential settlement.

MECHANISM

— Herstatt Risk

The risk that one party to an FX transaction completes its payment leg but the counterparty defaults before completing theirs. Atomic on-chain settlement eliminates this structurally.

RISK

— Real-Time FX vs T+2

Traditional FX settles T+2 (two business days after trade). On-chain settlement is near-instant — reducing counterparty risk, freeing capital and enabling 24/7 FX operations.

INNOVATION

— Liquidity Aggregator

A protocol routing FX trades across multiple DEX pools to achieve the best composite rate — the on-chain equivalent of a multi-bank FX aggregator.

INFRASTRUCTURE

— On-Chain FX Pricing

Exchange rates derived from DEX pool ratios rather than bank quotes. Transparent, manipulation-resistant (within AMM constraints) and available around the clock.

MECHANISM

— Multi-Currency Wallet

A wallet natively supporting multiple currency stablecoins — enabling instant on-chain FX conversion and cross-border payments without bank involvement.

TOOLS

— Liquidity Mining (FX Pools)

Incentivising users to supply liquidity to cross-currency pools by rewarding them with governance tokens — bootstrapping FX liquidity in early-stage Web3 markets.

DEFI

— Capital Efficiency in DeFi FX

Concentrated liquidity protocols (e.g. Uniswap v3) allow LPs to deploy capital within specific price ranges — dramatically improving capital efficiency for stable FX pairs.

INNOVATION

— Cross-Chain FX Bridge

Infrastructure enabling FX conversion simultaneously with asset transfer across chains — swapping USDC on Ethereum for EURI on another chain in a single atomic operation.

INFRASTRUCTURE

— Nostro/Vostro Account Disruption

Traditional banks maintain reciprocal accounts (nostro/vostro) to facilitate FX. Stablecoin rails replace this with shared on-chain liquidity pools — reducing trapped capital.

CONCEPT

— 24/7 FX Markets On-Chain

Unlike traditional FX markets (closed weekends, settlement cut-offs), on-chain FX operates continuously — critical for global payments and treasury operations across time zones.

INNOVATION

— FX Volatility Risk in Crypto Corridors

Even stablecoin pairs can experience temporary depegging — creating FX risk in corridors that assume stable 1:1 rates. Monitoring peg stability is essential for treasury operations.

RISK

CHAPTER 09

Systemic Risk & Market Structure

Contagion, interconnectedness and structural vulnerabilities in the Web3 financial ecosystem.

— Systemic Risk in DeFi

The risk that the distress or failure of one component of the DeFi ecosystem propagates through interconnected protocols, triggering cascading losses beyond the original point of failure. The Financial Stability Board (FSB, 2022) identifies DeFi's composability, leverage and liquidity mismatches as primary transmission channels for systemic stress.

CONCEPT

— Contagion Risk

The transmission of financial distress from one entity or protocol to others through shared exposures, collateral links or market psychology. Estimates of the total crypto market value erased in the weeks following the May 2022 Terra/LUNA collapse and the subsequent CeFi failures (Three Arrows Capital, Celsius, Voyager) range widely depending on methodology; figures in the hundreds of billions are commonly cited for the broader market drawdown, as distinct from the direct loss on Terra/LUNA itself (see below).

RISK

— Terra/LUNA Collapse

The May 2022 implosion of the UST algorithmic stablecoin and its LUNA backing token — a textbook on-chain bank run. Direct market value destroyed across UST and LUNA is most commonly put at approximately $40–45 billion within about a week, triggering contagion across CeFi and DeFi.

CASE-STUDY

— Bank Run On-Chain

A rapid mass withdrawal from a DeFi protocol or stablecoin triggered by a loss of confidence — structurally faster than traditional bank runs due to 24/7 markets and frictionless redemption.

RISK

— Liquidity Crisis On-Chain

A sudden collapse of available liquidity in DeFi markets — TVL evaporates, slippage spikes, liquidations cascade. Self-reinforcing feedback loops make recovery difficult.

RISK

— Too-Big-To-Fail in DeFi

The risk that a DeFi protocol becomes so deeply embedded in the ecosystem that its failure would be catastrophic — raising questions about implicit bailouts and systemic protection.

CONCEPT

— Interconnectedness Risk

The degree to which protocols share collateral, liquidity or composability dependencies — meaning a failure at one node propagates rapidly through the network.

RISK

— Counterparty Risk Without CCP

In traditional finance, central counterparties (CCPs) absorb counterparty risk. DeFi operates without CCPs — smart contract logic substitutes, but cannot fully replicate systemic protection.

CONCEPT

— Concentration Risk (Stablecoins)

The systemic risk from over-reliance on a small number of dominant stablecoins (USDT, USDC) — a failure or regulatory action against either could destabilise the entire DeFi ecosystem.

RISK

— Protocol Dependency Chain

The multi-layer dependencies between DeFi protocols — e.g. Protocol A uses Protocol B as collateral, which uses Protocol C for pricing. Failure propagates down the chain.

RISK

— CeFi Contagion (2022)

The cascade of centralised crypto firm failures (Celsius, Three Arrows Capital, FTX) — demonstrating that counterparty risk in CeFi transmitted to DeFi through shared collateral and market sentiment.

CASE-STUDY

— Collateral Spiral

A self-reinforcing collapse: falling asset prices trigger liquidations → liquidations increase supply → prices fall further → more liquidations. A classic dynamic in over-leveraged systems.

MECHANISM

— Oracle Manipulation Risk

Systemic risk from reliance on shared price oracles — manipulating the price feed can trigger mass liquidations across all protocols using that oracle.

RISK

— DeFi Liquidity Illusion

Apparent liquidity in DeFi is often fragile — high TVL can evaporate rapidly in a stress event as LPs withdraw simultaneously, leaving protocols unable to process redemptions.

RISK

— Governance Attack Risk

The risk of a malicious actor acquiring sufficient governance tokens to pass harmful proposals — e.g. draining treasury funds. A structural vulnerability in on-chain governance.

RISK

— Regulatory Contagion

The risk that a regulatory action against a single protocol, stablecoin issuer or exchange triggers a market-wide confidence crisis — even among unaffected participants.

RISK

— Stress Testing DeFi

Simulating extreme market conditions to assess protocol resilience — e.g. what happens if ETH drops 50% in one hour? A growing practice among risk-aware DeFi teams.

PROCESS

— Systemic Stablecoin Risk

The risk that a major stablecoin depegs or collapses — given stablecoins are collateral and settlement currency throughout DeFi, the systemic impact would be profound.

RISK

— Cross-Protocol Liquidation Cascade

When falling collateral values trigger liquidations across multiple protocols simultaneously — overwhelming DEX liquidity and amplifying price declines.

MECHANISM

— Market Structure Fragility

The structural characteristics of Web3 markets that increase systemic vulnerability: 24/7 operation, global accessibility, high leverage, composability and absence of circuit breakers.

CONCEPT

— Macroprudential Policy for DeFi

Emerging regulatory thinking on applying systemic risk oversight to DeFi — analogous to capital requirements and liquidity coverage ratios in traditional banking supervision.

REGULATION

CHAPTER 10

Fraud Typologies & Global Anti-Scam Regulation

How money actually gets stolen from people, and how regulators worldwide are starting to assign the bill.

— Pig Butchering Scam

A long-con investment fraud in which the scammer builds a personal or romantic relationship with the victim over weeks or months before introducing a fake crypto trading platform, encouraging escalating deposits, then disappearing with the funds. Named for the practice of 'fattening the pig before slaughter'. Frequently run out of scam compounds in Southeast Asia, often themselves staffed by trafficked forced-labour victims — a factor increasingly recognised in FATF and Interpol guidance.

SCAM

— Romance Scam

A scam in which the fraudster fabricates a romantic relationship, typically via dating apps or social media, to build trust before requesting money — for emergencies, travel, or investment 'opportunities'. Pig butchering is a specific, crypto-flavoured evolution of this typology.

SCAM

— Authorised Push Payment (APP) Fraud

Fraud in which the victim is deceived into authorising a payment themselves — impersonation of a bank, supplier, or a romance/investment scam — as opposed to unauthorised fraud where the account holder's credentials are stolen and used without their knowledge. Because the payment is technically 'authorised', it historically fell outside traditional fraud-liability rules, driving the wave of reimbursement regulation covered below.

THREAT

— Business Email Compromise (BEC)

A scam in which a fraudster impersonates a company executive or supplier via a compromised or spoofed email account to redirect a legitimate payment, often an invoice, to an account they control. One of the highest-value fraud typologies by loss per incident according to FBI IC3 data.

THREAT

— Money Mule

An individual who receives and transfers illicitly obtained funds through their own bank or crypto account on behalf of a fraud or laundering network, whether knowingly or after being recruited under a false pretext (e.g. a 'remote job'). A core layering mechanism for both APP fraud proceeds and crypto scam proceeds.

THREAT

— Smurfing / Structuring

Deliberately breaking a large transaction into multiple smaller transactions to stay under reporting or monitoring thresholds — a classic placement and layering technique in money laundering, applicable to both fiat and on-chain transfers.

THREAT

— SIM Swapping

An attack in which a fraudster fraudulently ports a victim's phone number to a SIM card they control, typically via social engineering of a telecom provider, in order to intercept SMS-based two-factor authentication and take over banking or crypto exchange accounts.

THREAT

— Account Takeover (ATO)

Unauthorised access to and control of a victim's existing account, obtained via phishing, credential stuffing, malware, or SIM swapping, used to move funds or crypto assets out before the victim notices.

THREAT

— Synthetic Identity Fraud

Fraud that combines real and fabricated personal data (e.g. a real national ID number with a fictitious name) to create a new, non-existent identity used to open accounts and build credit before 'busting out'. Harder to detect than classic identity theft because there is no real victim to report it.

THREAT

— Recovery Scam

A secondary scam targeting people who have already lost money to fraud, in which the fraudster poses as a lawyer, regulator or 'asset recovery specialist' offering to retrieve the lost funds for an upfront fee.

SCAM

— UK PSR APP Fraud Reimbursement Requirement

A UK Payment Systems Regulator rule in force since 7 October 2024 requiring payment service providers on Faster Payments and CHAPS to reimburse in-scope victims of authorised push payment fraud in most cases, with the cost split equally between the sending and receiving firm, capped at GBP 85,000 per claim. The first mandatory, industry-wide APP fraud reimbursement regime of its kind.

REGULATION

— EU PSD2 / PSD3 & PSR

The EU's Payment Services Directive framework, currently being recast as PSD3 and a directly applicable Payment Services Regulation (PSR), which is expected to introduce stronger authentication, IBAN/name-matching checks, and expanded fraud-prevention and information-sharing obligations for payment service providers, moving the EU closer to a UK-style liability framework for APP-type fraud.

REGULATION

— Australia Scams Prevention Framework (SPF)

An Australian federal law (Scams Prevention Framework Act 2025), commenced 21 February 2025, imposing statutory prevent-detect-disrupt-respond-report obligations on regulated banks, telecommunications providers and digital platforms, with civil penalties of up to AUD 50 million per contravention. Obligations are phased in, with most substantive duties applying from 2026–2027.

REGULATION

— Singapore Shared Responsibility Framework (SRF)

A framework in force since December 2024, overseen jointly by the Monetary Authority of Singapore (MAS) and the Infocomm Media Development Authority (IMDA), allocating scam losses between banks and telecommunications operators based on whether each met specific anti-phishing duties. Unlike the UK and Australian regimes, the SRF is scoped to phishing scams with a digital nexus and explicitly excludes scams where the victim authorised the payment themselves.

REGULATION

— Regulation E / EFTA (US)

The US Electronic Fund Transfer Act and its implementing Regulation E give consumers strong protection against unauthorised electronic transfers, but — unlike the UK and emerging EU/Australian regimes — generally do not require reimbursement where the consumer was tricked into authorising the transfer themselves, leaving APP-style fraud largely uncovered in the US market.

REGULATION

— FinCEN (US)

The US Financial Crimes Enforcement Network, the primary US AML/CFT regulator and financial intelligence unit, responsible for Bank Secrecy Act rulemaking, SAR collection, and — following the GENIUS Act — AML supervision of payment stablecoin issuers.

REGULATION

— FBI IC3 (Internet Crime Complaint Center)

The US public reporting portal and data clearinghouse for cyber-enabled fraud, including romance scams, BEC, and crypto investment fraud. Its annual Internet Crime Report is a primary source for fraud-typology loss statistics.

INTELLIGENCE

— Europol EMPACT

The EU's Multidisciplinary Platform Against Criminal Threats — a coordinated law-enforcement framework under which EU member states run joint operations against organised fraud typologies, including romance/investment scams and money-mule networks.

REGULATION

— Confirmation of Payee (CoP)

A name-checking service that verifies whether the account name a payer enters matches the actual account holder before a transfer is sent, designed to catch misdirected or fraudulent payments, including APP fraud, before money leaves the account. Mandatory for UK payment providers on Faster Payments and CHAPS; the EU's equivalent verification-of-payee obligation is being phased in as part of the instant-payments and PSD3 agenda.

REGULATION

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